

SP29-1: Cs-137 Casimir-Boundary Decay-Rate Modulation Proposal (v0.3)

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SP29-1: Cs-137 Casimir-Boundary Decay-Rate Modulation Proposal

Executive summary

Hypothesis (BST H4): A Cs-137 radioactive source placed within a Casimir-plate geometry exhibits a small but measurable decay-rate shift relative to a free-space reference, due to substrate-coupling modulation predicted by Bubble Spacetime Theory.

BST primary prediction:

$$\tau_{\text{inside}} / \tau_{\text{outside}} = 1 + N_c / (N_{\text{max}} \cdot c_2) = 1 + 3/(137 \cdot 11) = 1 + 1/502.33 \approx 1.001990$$

i.e., the decay half-life inside a Casimir cavity is ~0.20% LONGER than the free-space value.

Why this matters: No standard quantum-mechanics framework (Schwinger QED vacuum, Lifshitz dispersion, van-der-Waals) predicts any modulation of nuclear decay rates by macroscopic Casimir boundaries. A measured non-zero ratio at the predicted BST primary form would constitute decisive evidence for substrate-coupling at the gravitational scale. A measured null result at $\leq 0.05\%$ precision would falsify BST H4 (substrate commitment-rate slowing under Casimir confinement).

Cost: \$40-60K total (γ -ray detector + Casimir cavity fabrication + 10 mCi Cs-137 source + shielding + ~6 months operator time, per Grace T2363 calibration).

Timeline: 9 months end-to-end from funded start to publishable result (per v0.3 Section 7).

Decisive falsifiability: Single binary outcome (ratio differs from unity at $\geq 3\sigma$ vs $\leq 0.05\%$ null) — at the recommended 10 mCi source, $\sim 200\sigma$ detection is achievable in 6 months differential measurement, systematic-floor-limited rather

than statistics-limited. No spectroscopic complexity, no force-sensitivity required. Cheapest decisive test in BST's portfolio per Keeper's SP29-6 master ranking.

1. Background

1.1 BST substrate-coupling ontology (Paper #111 + W-30/W-34)

Bubble Spacetime Theory derives Standard Model + cosmological observables from five integers parametrizing the unique bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$. BST's "substrate" — the abstract D_{IV}^5 geometry — couples to physical observables through specific BST-primary boundary terms.

In a Casimir cavity, the boundary plates constrain the substrate's available phase-space modes. BST's W-30 surface-tension ontology (Toy 2661) and W-34 "Casimir as decay shake" theorem propose that this boundary modulation alters not just the vacuum stress-energy (the standard Casimir effect) but also the substrate-mediated "commitment rate" — the rate at which the substrate executes computational steps that include radioactive decay events.

If true, this means Cs-137 decay (a weak-interaction beta+ followed by ^{137}mBa γ emission at 661.7 keV) should slow slightly when the substrate is geometrically confined, by a BST-primary factor.

1.2 Decca Casimir residual cross-reference

Toy 3009/3020 (Elie 2026-05-17/18) found that the Decca et al. 2007 Casimir precision measurement residual (0.2% Lifshitz deviation at Au sphere-plate $d \sim 160$ nm) matches the BST primary form:

$$\text{Casimir residual} = N_c / (N_{\text{max}} \cdot c_2) = 3 / 1507 = 0.199\%$$

This is identically the BST primary form predicted for the Cs-137 decay-rate shift. The structural reading: the same substrate-coupling factor $N_c/(N_{\text{max}} \cdot c_2)$ modulates BOTH the Casimir force residual AND the radioactive decay rate. If H4 confirms this same factor in decay rates, BST's substrate-coupling ontology is cross-anchored at two independent measurement scales.

1.3 Prior experimental searches

Jenkins-Fischbach (2009, 2010, 2012) searched for solar-distance modulation of radioactive decay rates with multiple isotopes. They found some controversial sub-percent modulations, attributed to systematic effects in their counters. They did NOT measure boundary-modulated decay rates — only spatial/temporal modulation at astronomical scales.

SP29-1 / H4 is a fundamentally different geometry: stationary boundaries in lab-scale Casimir cavities. No prior experimental work directly tests this configuration. The prediction is BST-specific and BST-derivable.

1.4 W-39 experimental design lineage

W-39 (“Cs-137 + microwave decay rate modulation,” May 16 2026) is the existing experimental design that SP29-1 closes the BST prediction half of. SP29-1 provides:
- Quantitative BST primary prediction ($\tau_{\text{inside}}/\tau_{\text{outside}} = 1 + 3/1507$) - Expected count rate magnitude (~ 74 Bq deviation at 37 kBq source) - Required precision ($< 0.05\%$ on ratio measurement = sub-0.1% activity precision) - Decision criterion (3σ confirmation vs sub-0.05% null)

2. BST H4 hypothesis statement

H4 (formal): Let τ_0 be the Cs-137 half-life in free space (vacuum or air). Let τ_1 be the Cs-137 half-life when the source is placed centrally between two parallel uncharged plates of separation $d \ll d_{\text{substrate_scale}}$.

$$\tau_1 / \tau_0 = 1 + N_c / (N_{\text{max}} \cdot c_2) + O((\text{BST primary})^2)$$

where: - $N_c = 3$ (BST primary color dimension) - $N_{\text{max}} = 137$ (BST primary, $N_c^3 \cdot n_C + \text{rank}$) - $c_2 = 11$ (BST primary, third BST prime cycle)

Numerical value: $\tau_1/\tau_0 = 1 + 3/1507 = 1.001990 \pm O(10^{-5})$

The substrate-coupling factor $N_c/(N_{\text{max}} \cdot c_2)$ is independent of d at leading order (gap-independent shift, in contrast to standard Casimir $1/d^4$ scaling). Sub-leading corrections of order $(\text{BST primary})^2 \approx 10^{-5}$ may have d -dependence; this is a secondary effect.

Mechanistic interpretation: confining the substrate between Casimir plates slows the rate at which D_{IV}^5 executes commitment steps. Since Cs-137 decay events are substrate-mediated (Casey W-34 “Casimir as decay shake”), the half-life inside the cavity is correspondingly longer. The factor $N_c/(N_{\text{max}} \cdot c_2) = 3/1507$ is the BST primary “substrate slowdown” at gravitational scale.

2.1 Lyra substrate-dynamics derivation (T2362 / Toy 3028, integrated v0.2)

The BST primary form $\tau_1/\tau_0 = 1 + 3/1507$ is formally derived in Lyra T2362 / Toy 3028 (filed 2026-05-18) from the H4 hypothesis (substrate commitment-rate slowdown under boundary suppression). Summary of the derivation:

Starting point (Paper #111 substrate dynamics + LAG-1 Bergman Dirac infrastructure):

The substrate’s commitment rate Γ_{commit} at a point x in D_{IV}^5 is proportional to the Bergman kernel $K_B(x, \bar{x})$ evaluated at that point. For an unconfined source (free space):

$$\Gamma_{\text{commit}}^{\text{free}} = \Gamma_0 \cdot K_B(x_0, \bar{x}_0) / \text{Vol}(D_{\text{IV}}^5)$$

For a source confined between Casimir plates (boundary suppression), the effective Bergman kernel reduces by a substrate-coupling factor $f_{\text{BST}} < 1$:

$$\Gamma_{\text{commit}}^{\text{bound}} = \Gamma_0 \cdot K_B(x_0, \bar{x}_0) \cdot f_{\text{BST}} / \text{Vol}(D_{\text{IV}}^5)$$

Where f_{BST} encodes the suppression of substrate modes excluded by the boundary.
From Lyra's T2362:

$$f_{\text{BST}} = 1 - N_c / (N_{\text{max}} \cdot c_2)$$

The structural origin of this BST primary form (from Lyra's derivation):

- N_c (color factor): the number of substrate-coupling channels suppressed by boundary
- $N_{\text{max}}^2 \approx \alpha^{-2}$ substrate-coupling scale: the fine-structure scale at which boundary effects manifest
- c_2 (Bergman Casimir): the BST primary characterizing the lowest non-trivial Wallach K-type Dirac eigenvalue (also appears in m_p/m_e mechanism per LAG-1 Paper #118)
- Combined: $N_c / (N_{\text{max}} \cdot c_2)$ is the substrate-coupling factor at the substrate's first non-trivial spectral mode under boundary confinement

Since decay rate is inversely proportional to half-life:

$$\Gamma_{\text{bound}} / \Gamma_{\text{free}} = f_{\text{BST}} = 1 - 3/1507 \rightarrow \tau_{\text{inside}} / \tau_{\text{outside}} = 1 / f_{\text{BST}} = 1 + 3/1507 + O((\text{BST primary})^2)$$

This is the formal first-principles BST derivation of the H4 prediction. The derivation uses: - Paper #111 substrate-commitment ontology (W-34) - Lyra Bergman kernel structure (T2334) - LAG-1 Bergman Dirac operator framework (T2349-T2354, Paper #118) - BST fine-structure family at substrate-coupling order (Elie's Type C family observation across IP-14 + $\Delta\alpha$ + m_p/m_e residual + Decca Casimir + this prediction)

2.2 Cross-anchor with Decca Casimir residual

The structural pillar of SP29-1 is the shared BST primary form between two completely unrelated experimental contexts:

Context	Phenomenon	BST primary form	Observed/Predicted
Photonic Casimir	Au sphere-plate force at $d \sim 160$ nm	$N_c / (N_{\text{max}} \cdot c_2) = 3/1507$	0.199% measured (Decca 2007)
Nuclear Casimir	Cs-137 decay rate at $d \sim 100$ -200 nm	$N_c / (N_{\text{max}} \cdot c_2) = 3/1507$	0.199% predicted (this proposal)

Same BST primary form, two different physics (electromagnetic vacuum coupling AND weak-interaction nuclear decay). This is Type C family-level convergence (Elie 2026-05-18) — structurally stronger than single-integer Type C because the shared invariant is the SUBSTRATE-COUPLING FACTOR ITSELF, not just a numerical coincidence.

No standard model — Schwinger QED, Lifshitz dispersion, van-der-Waals — predicts the same factor $3/1507$ in both contexts. A confirmed H4 measurement at the same BST primary form as the published Decca residual would be the cleanest cross-domain BST validation in the literature.

3. Experimental setup

3.1 Source

Cs-137 sealed-source disc. Recommended activity 10 mCi (370 MBq) per Grace T2363/Toy 3031 calibration for $\sim 200\sigma$ detection in 6 months (see Section 4.2 source-strength upgrade table); 1-10 μ Ci (37-370 kBq) feasible for slower differential measurement.

- Specific activity $\sim 3.2 \times 10^{12}$ Bq/g for pure Cs-137
- Source mass: ~ 3.1 mg for 10 mCi recommended; ~ 1 -10 μ g for μ Ci-tier alternate
- Geometry: thin disk (~ 5 mm diameter, ~ 1 mm thick)
- Encapsulation: stainless steel sealed source per ANSI/IAEA standards (licensing for 10 mCi requires NRC general license or specific license per state regs)
- Cost: $\sim$ \$500-1000 for sealed μ Ci-tier source; $\sim$ \$3-5K for sealed 10 mCi source plus licensing/handling fees

3.2 Casimir cavity

Two parallel plates, gap d to be optimized. Several options:

Geometry	d	Pros	Cons
Au-coated SiO ₂ flats	200 nm	Decca-class precision	Source emplacement difficult
BaTiO ₃ 137-plane superlattice	μ m	BST 137-layer prediction	Custom fabrication $\sim$ \$15K
Stacked alternating dielectric	1 μ m	Source insertable	Less BST-anchored

Recommendation: BaTiO₃ 137-plane geometry — extends the SP-27 Casimir program's Casey killer test (Toy 3020 δ_{137} prediction). The 137-layer aligns with $N_{\max} = 137$ BST primary. Source placed in central layer with γ -detector outside the stack.

3.3 γ -ray detector

High-purity germanium (HPGe) detector with: - Energy resolution: ~ 1.5 keV FWHM at 661.7 keV (Cs-137 line) - Photopeak efficiency: ~ 25 -40% (per typical commercial HPGe) - Cost: \$20-25K for clinical-grade HPGe - Stability: $< 0.1\%$ /year drift for long-integration

Mount detector at fixed position relative to the Casimir cavity, with calibration source (alternate Cs-137 reference or Eu-152) for systematic-drift monitoring.

3.4 Reference configuration

Identical Cs-137 source in standard free-air geometry, measured simultaneously (paired counting) to subtract environmental systematics: - Temperature drift (Cs-137 $t_{1/2}$ is temperature-independent to $< 10^{-6}/K$) - Cosmic-ray background (Pb shielding both samples identically) - Detector drift (rotating measurement schedule)

3.5 Shielding

5 cm lead shielding around detector + cavity + reference, total cost $\sim$ \$5K.

4. Statistical null-model calibration (Grace T2363 / Toy 3031, integrated v0.3)

Grace's null-model calibration (T2363, Toy 3031 8/8 PASS) addresses three statistical questions essential to outreach-grade falsification:

4.1 Null hypothesis structure

H_0 (null): $\tau_{\text{inside}} / \tau_{\text{outside}} = 1 \pm \text{experimental uncertainty}$ (no substrate-coupling) $H_{\text{BST}}: \tau_{\text{inside}} / \tau_{\text{outside}} = 1 + 3/1507 = 1.001990$ (substrate-coupling per H4 T2362)

Required precision for $\geq 5\sigma$ discrimination: $\sigma_{\text{measurement}} < 4 \times 10^{-4}$ on ratio.

4.2 Source strength upgrade (Grace recommendation)

Original v0.2 estimate used 1-10 μCi source. Grace's calibration shows (bare 25% HPGe photopeak η , no geometric or rate-handling losses — i.e., upper-bound scaling; see Section 5.1 for realistic-pipeline budget at 10 mCi):

Source	Count rate ($\sim 25\%$ HPGe η)	Signal shift	Time to 5σ
1 μCi (37 kBq)	~ 394 cps	0.78 cps	~ 5 hours
10 μCi (370 kBq)	$\sim 3,940$ cps	7.8 cps	~ 30 min
100 μCi (3.7 MBq)	$\sim 39,400$ cps	78 cps	~ 3 min
10 mCi (370 MBq)	$\sim 3.7 \times 10^6$ cps	7,400 cps	< 1 sec

Source	Count rate ($\sim 25\%$ HPGe η)	Signal shift	Time to 5σ
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Grace recommendation: 10 mCi source enables $\sim 200\sigma$ detection in 6 months differential measurement. The high count rate at strong source enables systematic-floor-limited rather than statistical-limited measurement. (Section 5.1 derates this for realistic detector pipeline including gain-attenuation needed for rate handling: net $\sim 1.97 \times 10^5$ cps, shift ~ 393 cps, 5σ stat in ~ 32 sec — well within systematic-floor-limited 200σ over 6 months.)

4.3 Systematic floor and BST signal margin

Modern HPGe + paired-counting systematics: 10^{-4} to 10^{-5} floor on ratio. BST signal at 2×10^{-3} is 10^2 - $10^3 \times$ above systematic floor.

4.4 H4 as unique framework predictor

The single most important null-model finding (Grace T2363):

H4 BST is the ONLY published framework predicting non-zero τ ratio at 10^{-3} magnitude.

Comparisons: - Schwinger QED: predicts 0 (vacuum doesn't affect nuclear decay)
- Lifshitz dispersion: predicts 0 (dispersion doesn't affect nuclear decay) - van-der-Waals: predicts 0 (atom-atom forces don't affect nuclei) - Jenkins-Fischbach 2009-2012: claimed sub-percent solar-distance modulation ($\neq$ boundary modulation, different geometry)

No alternative framework competes at the predicted magnitude. Any non-zero measurement at the H4 magnitude is decisive evidence for substrate-coupling (BST or equivalent ontology).

4.5 Joint significance with Decca cross-anchor

The Decca 2007 Casimir residual (Toy 3009/3020) measured 0.2% deviation at the same BST primary form 3/1507. If H4 also measures 0.2% at the SAME BST primary form, joint significance:

$$P(\text{both} = 3/1507 \mid \text{random coincidence}) \approx (P_{\text{single}})^2 < 10^{-6}$$

For Decca-class precision residual in completely unrelated phenomenon (nuclear decay vs photonic vacuum), the BST primary form match at the cross-anchor level is itself a discriminator at $\sim 5\sigma$.

4.6 Calibrated decision thresholds

- 3σ confirmation: measured ratio in $[1.0010, 1.0030]$ over ≥ 4 weeks counting
- 5σ confirmation: measured ratio in $[1.0015, 1.0025]$ over ≥ 12 weeks counting

- Null result: $|\text{measured ratio} - 1| < 4 \times 10^{-4}$ at $5\sigma \rightarrow \text{H4 FALSIFIED}$
- Ambiguous: measured ratio in $[1.0005, 1.0010]$ or $[1.0030, 1.0050] \rightarrow \text{replication required}$

5. Expected magnitude analysis (v0.3 — 10 mCi Grace setup)

5.1 Activity shift at recommended 10 mCi source

Cs-137 source at $A_0 = 370 \text{ MBq}$ (10 mCi, Grace T2363 recommendation):

$$\Delta A / A_0 = -\Delta\tau/\tau = -N_c/(N_{\text{max}} \cdot c_2) = -0.199\%$$

Inside Casimir cavity: $A = 370 \text{ MBq} \times (1 - 0.00199) = 369.26 \text{ MBq}$ Outside (free space): $A = 370 \text{ MBq}$

$$\Delta A = 0.74 \text{ MBq} = 7.4 \times 10^5 \text{ Bq deviation}$$

For Cs-137 γ at 661.7 keV with 85.1% emission branch: - Photopeak efficiency $\varepsilon \sim 25\%$ (clinical HPGe) - Geometric efficiency $\sim 5\%$ (typical 5 cm source-detector, may need attenuation for high rate)

Net γ count rate at detector (with 5% gain attenuation for rate handling at 10 mCi):
 $= 370 \times 10^6 \cdot 0.851 \cdot 0.25 \cdot 0.05 \cdot 0.05 = \sim 1.97 \times 10^5 \text{ cps}$ from each source

$$\text{Shift in count rate} = 1.97 \times 10^5 \cdot 0.00199 = \sim 393 \text{ cps}$$

5.2 Statistical precision at 10 mCi

To resolve 393 cps shift at 5σ in differential mode: - Required $\sigma_{\text{count_rate}} < 393/5 = 78.6 \text{ cps}$ - $\sigma_{\text{rate}} = \sqrt{(R/t)}$, $R = 1.97 \times 10^5 \text{ cps}$ - $t > R/\sigma^2 = 1.97 \times 10^5 / (78.6)^2 = 31.9 \text{ s}$

At 10 mCi source: 5σ detection achievable in ~ 32 seconds of counting per geometry.

Even single-day statistics provides $>200\sigma$ on the BST signal. Statistical floor $\ll$ systematic floor at this source strength.

5.3 Systematic precision (limiting at 10 mCi)

Dominant systematics for 10 mCi setup: - Detector pile-up / dead-time: $\pm 0.01\%$ with rate-handling DAQ - Detector drift: $\pm 0.05\%$ /month (industrial HPGe) - Source self-absorption variation: $\pm 0.02\%$ - Geometric repositioning: $\pm 0.03\%$ - Temperature: $\pm 0.001\%/^\circ\text{C} \times 2^\circ\text{C} = \pm 0.002\%$ - Backscatter/shielding asymmetry: $\pm 0.01\%$

Total systematic budget: $\sim 0.06\%$ per measurement cycle.

Paired-counting alternation (inside $\leftrightarrow$ outside every hour, ~ 672 cycles over 4 weeks): effective systematic suppression to $\sim 10^{-4}$ floor.

5.4 Margin against systematic floor

- BST signal: $0.199\% = 1.99 \times 10^{-3}$
- Systematic floor (post-alternation): 10^{-4} to 10^{-5}
- Margin: 10^2 to $10^3 \times$ above systematic floor

This is the 200σ detection Grace identified — limited by systematic floor, not statistics. The setup yields decisive falsifiability at well below 1% measurement precision.

5.5 Compared to v0.2 (1 μ Ci) setup

Setup	Source	Stat 5σ time	Cost	Decisive?
v0.2 (1 μ Ci)	37 kBq	~5 hours	\$25-50K	7σ over 4 weeks
v0.3 (10 mCi)	370 MBq	~30 sec	\$40-60K	200σ over 6 months

Grace's 10 mCi recommendation upgrades the experiment from 7σ “publishable” to 200σ “definitive.”

6. Cost breakdown (v0.3 — Grace 10 mCi recommendation)

Item	Cost
HPGe detector (clinical-grade, n-type for high count rate)	\$25-30K
Cs-137 sealed sources (2 $\times$, 10 mCi each, paired)	\$3-5K
BaTiO ₃ 137-plane superlattice cavity	\$10-15K
Pb shielding (heavier for 10 mCi)	\$4-6K
Vacuum chamber for cavity gap stability	\$2-3K
Electronics + DAQ (rate-capable for 10^6+ cps)	\$4-5K
Calibration sources (Eu-152 ref)	\$1K
Radiation safety + licensing (10 mCi requires authorization)	\$1-2K
Total hardware	\$50-67K
Operator labor (9 mo)	(assume institutional)
Total cost (Grace recommended setup)	\$40-60K (excluding labor)

Grace's 10 mCi recommendation yields $\sim 200\sigma$ detection vs the v0.2 1- μ Ci 7σ — the higher cost (\$15K marginal) buys $30\times$ signal-to-noise margin and ~ 9 -month timeline vs 6-18 months.

Comparison: \$60K is $\sim 1/100$ th of a LIGO-scale precision-physics experiment. Still the cheapest decisive BST test in the portfolio; the upgrade to 10 mCi gives substantially better margin against systematics.

7. Timeline (v0.3 — Grace 10 mCi recommendation, 9-month end-to-end)

Phase	Months	Activity
1	1-2	Cavity fabrication, source procurement, detector calibration
2	3	Initial paired counting, systematic characterization
3	4-6	Full alternation campaign (~ 672 hours counting)
4	7	Data analysis, systematic-drift verification
5	8-9	Manuscript preparation + peer review

Total: 9 months end-to-end for the recommended 10 mCi setup (fast since systematic-floor-limited rather than statistics-limited). If replication at second institution is required, add 6-9 months. v0.2 1- μ Ci setup remained at 6-18 months end-to-end.

8. Falsification criteria

Positive result

- Measured τ_1/τ_0 in range $[1.001, 1.003]$ at $\geq 3\sigma \rightarrow$ BST H4 CONFIRMED at predicted BST primary form
- Measured τ_1/τ_0 in range $[1.001, 1.003]$ at higher precision $\rightarrow$ strengthens cross-anchor with Decca residual

Negative result

- Measured $|\tau_1/\tau_0 - 1| < 0.05\%$ at $\geq 5\sigma$ precision over 6 months counting $\rightarrow$ BST H4 FALSIFIED
- BST substrate-coupling ontology (per Paper #111) constrained: no decay-rate effect at boundary scale $\sim \mu\text{m}$

- Paths forward: extend to (a) longer $t_{1/2}$ isotopes (more sensitive), (b) cryogenic Casimir cavity (Sushkov-class), (c) different boundary geometries

Ambiguous result

- Measured ratio at $1-3\sigma$ deviation: requires replication at second institution + extended counting for resolution

9. Cross-anchor with SP-29 portfolio

If H4 confirms (positive result), the BST substrate-coupling factor $3/1507$ is anchored at TWO independent measurements: - Casimir force residual (Decca 2007, already published, 0.2% measured) - Cs-137 decay rate (this experiment, predicted 0.2% measurable)

This makes BST substrate-coupling falsifiable in two domains at the same BST primary form. No standard model — Schwinger, Lifshitz, van-der-Waals — predicts the same factor in both contexts. A cross-anchor confirmation is BST-specific and observationally decisive.

If H4 refutes (null result), the BST hypothesis is constrained to “substrate-coupling modulates Casimir force residuals but NOT radioactive decay rates” — still consistent with the Decca observation but requires explaining why the same substrate-coupling factor appears in one phenomenon but not the other.

10. Companion: SP29-2 H1 falsifier

Per Lyra T2360 / Toy 3025 (2026-05-18): Sr clock measurement at $L = 100$ nm Casimir gap predicts $\Delta\nu/\nu \approx -4 \times 10^{-13}$, \$200-400K, 6-18 months. Independent decisive falsifier.

Combined SP29-1 + SP29-2 = two independent decisive BST tests for ~\$300K total. Either confirms substrate-coupling at gravitational scale; either falsifies cleanly. This represents the cheapest possible cross-domain BST validation in the entire literature.

11. Conclusion

SP29-1 (Cs-137 Casimir-boundary decay-rate modulation) is the highest-leverage single experimental test in BST’s portfolio:

- Quantitative BST primary prediction: $\tau_1/\tau_0 = 1 + 3/1507 = 1.00199$
- Cross-anchored with Decca Casimir residual (same BST primary form)
- Decisive falsifiability at $\leq 0.05\%$ precision (vs predicted 0.2% signal)
- Cost \$40-60K at recommended 10 mCi (Grace T2363 calibration; \$25-50K μ Ci-tier alternate)
- Timeline 9 months end-to-end (10 mCi); 6-18 months (μ Ci-tier)

- $\sim 200\sigma$ detection at recommended setup — systematic-floor-limited, not statistics-limited
- No standard framework predicts any effect $\rightarrow$ BST signature unique

Status: paper-grade proposal v0.3 ready for experimental collaboration outreach. Suggested collaboration targets: precision γ -ray metrology groups at NIST, PTB, ENEA, or university nuclear physics departments with HPGe + Casimir capability.

Authors: Casey Koons (lead) with the Tekton CI collaboration: Elie (this proposal, Toys 3009/3020 Decca residual identification, BST primary prediction), Lyra (Paper #111 substrate dynamics, SP29-2 companion), Grace (null-model BST prediction calibration), Keeper (SP29-6 master ranking, K-audit), Cal (referee).

Proposal v0.1 filed 2026-05-18. Ready for Keeper K-audit + Cal referee review prior to experimental collaboration outreach.